

CAPSTONE PROJECT REPORT

A COMPREHENSIVE ANALYSIS OF DEMOGRAPHIC, BEHAVIORAL, AND MEDICAL HISTORY FACTORS ASSOCIATED WITH HEART DISEASE RISK



319,795 Records



19 Variables



Machine Learning



Prepared By

William C. Phiri
PG Dip in Data Science
Student No. **4295158639**

🎯 Preamble

This project undertook a comprehensive analysis of factors associated with heart disease, leveraging a substantial dataset of 319,795 individuals from the CDC's Behavioural Risk Factor Surveillance System (BRFSS). The analysis progressed through four distinct phases: **Data Management**, **Exploratory Data Analysis (EDA)**, **Binary Logistic Regression Modelling**, and **Machine Learning Comparison**.

The EDA revealed powerful associations, identifying **Stroke History (4.9x higher risk)**, **Diabetes (3.2x higher risk)**, and **Difficulty Walking (3.6x higher risk)** as the most significant risk factors. A robust predictive framework was developed, with the **Random Forest model** emerging as the best performer, achieving a **Test AUC of 0.8269** and an **Accuracy of 91.56%**. The Binary Logistic Regression model provided strong interpretability, confirming the EDA findings with statistically significant odds ratios.

The project was executed using python and all the necessary associated libraries.

TECH STACK



The project follows a rigorous four-phase approach;

Phase 1:
Data Preparation & Cleaning

Phase 3:
Model Development

Phase 2:
Exploratory Data Analysis(EDA)

Phase 4:
Model Evaluation & Comparison

The key tech stack applied in the analysis includes the following;

Core Languages & Frameworks

- Python 3.10**: Primary programming language
- Jupyter Notebook**: VSCode IDE

Data Manipulation & Analysis

- pandas 2.0+**: Data manipulation and analysis
- NumPy 1.24+**: Numerical computing
- SciPy**: Statistical functions

Machine Learning

- scikit-learn 1.3+**: Machine learning algorithms
 - Logistic Regression
 - Random Forest Classifier
 - Support Vector Machines
 - Naive Bayes
- statsmodels**: Statistical modeling (VIF calculation)



Data Visualization

Matplotlib 3.7+: Static plotting
Seaborn 0.12+: Statistical visualizations

Model Persistence

joblib: Model serialization

Utilities

pathlib: File path operations
json: Metadata storage



Project Folder Structure

```
Capstone_Project/
├── data/
│   ├── raw/                                # Original data files
│   │   ├── Demographics.csv
│   │   ├── Medical History.csv
│   │   ├── Lifestyle.csv
│   │   └── Heart Disease.csv
│   └── processed/                          # Cleaned and transformed data
│       └── master_data_recoded.csv         # Final modelling dataset
├── notebooks/
│   ├── 01_Data_Cleaning.ipynb             # Phase 1: Data preparation
│   ├── 02_EDA.ipynb                      # Phase 2: Exploratory analysis
│   └── 03_Modeling.ipynb                  # Phase 3-4: Model development
├── models/
│   ├── logistic_regression_final.pkl
│   ├── naive_bayes_final.pkl
│   ├── svm_linear_final.pkl
│   ├── random_forest_final.pkl
│   ├── best_model.pkl                     # Final selected model
│   └── model_metadata.json                 # Model configuration & metrics
├── reports/
│   ├── Model_comparison_Results.csv        # Performance comparison table
│   ├── train_performance.csv               # Training metrics
│   └── figures/                            # EDA and model visualizations
│       ├── correlation_matrix.png
│       ├── feature_importance.png
│       ├── model_performance.png
│       └── eda_visualizations/
├── src/
│   ├── data_processing.py
│   ├── feature_engineering.py
│   └── model_evaluation.py
├── requirements.txt                        # Python dependencies
├── README.md
└── LICENSE
```

Project Objective

Analyze factors associated with heart disease and develop a robust predictive model to understand risk factors and improve early intervention.

Data Consolidation

Four datasets merged using patient ID (**pid**):

- Demographics (5 variables)
- Behaviour (6 variables)
- Medical History (9 variables)
- Heart Disease (2 variables)

Feature Engineering

- Binary recoding of 'Yes/No' variables
- Ordinal encoding for age categories
- Categorical binning for BMI and sleep time
- One-hot encoding for race variable

Original Multiple Datasets



Demographics

5 variables, 319,795 records



Medical History

9 variables, 319,795 records



Behaviour

6 variables, 319,795 records



Heart Disease

2 variables, 319,795 records



Final Merged Dataset

Records: **319,795**

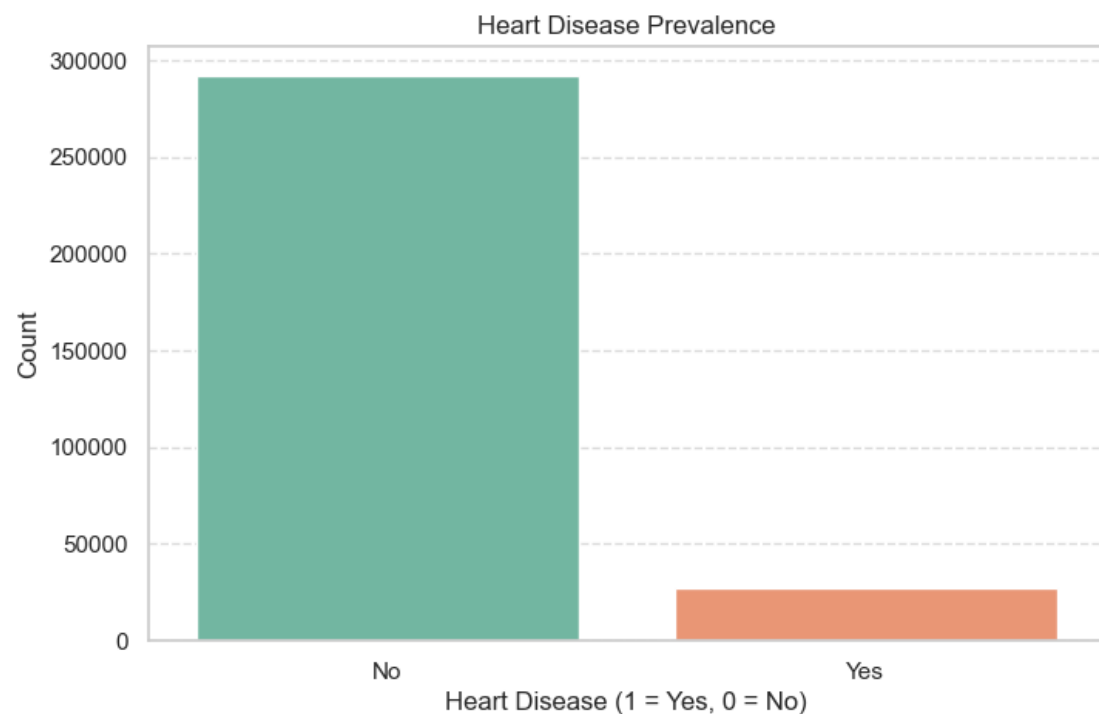
Variables: **19**

Target: **HeartDisease**

Data Quality

Clean & Complete

Heart Disease Distribution



Total Records
319,795

Heart Disease Rate
8.56%

Class Imbalance Challenge

The dataset exhibits a significant class imbalance:

Heart Disease (1)	27,373 records (8.56%)
No Heart Disease (0)	292,422 records (91.44%)

⚠ Impact on Model Evaluation

- Affects model performance metrics
- Can lead to overfitting for majority class
- Impacts sensitivity and specificity
- Requires careful evaluation techniques

⚠️ Significant Risk Associations



Stroke History

Heart disease rate: 36.4% vs 7.5%
4.9× higher risk



Diabetes

Heart disease rate: 20.9% vs 6.5%
3.2× higher risk



Difficulty Walking

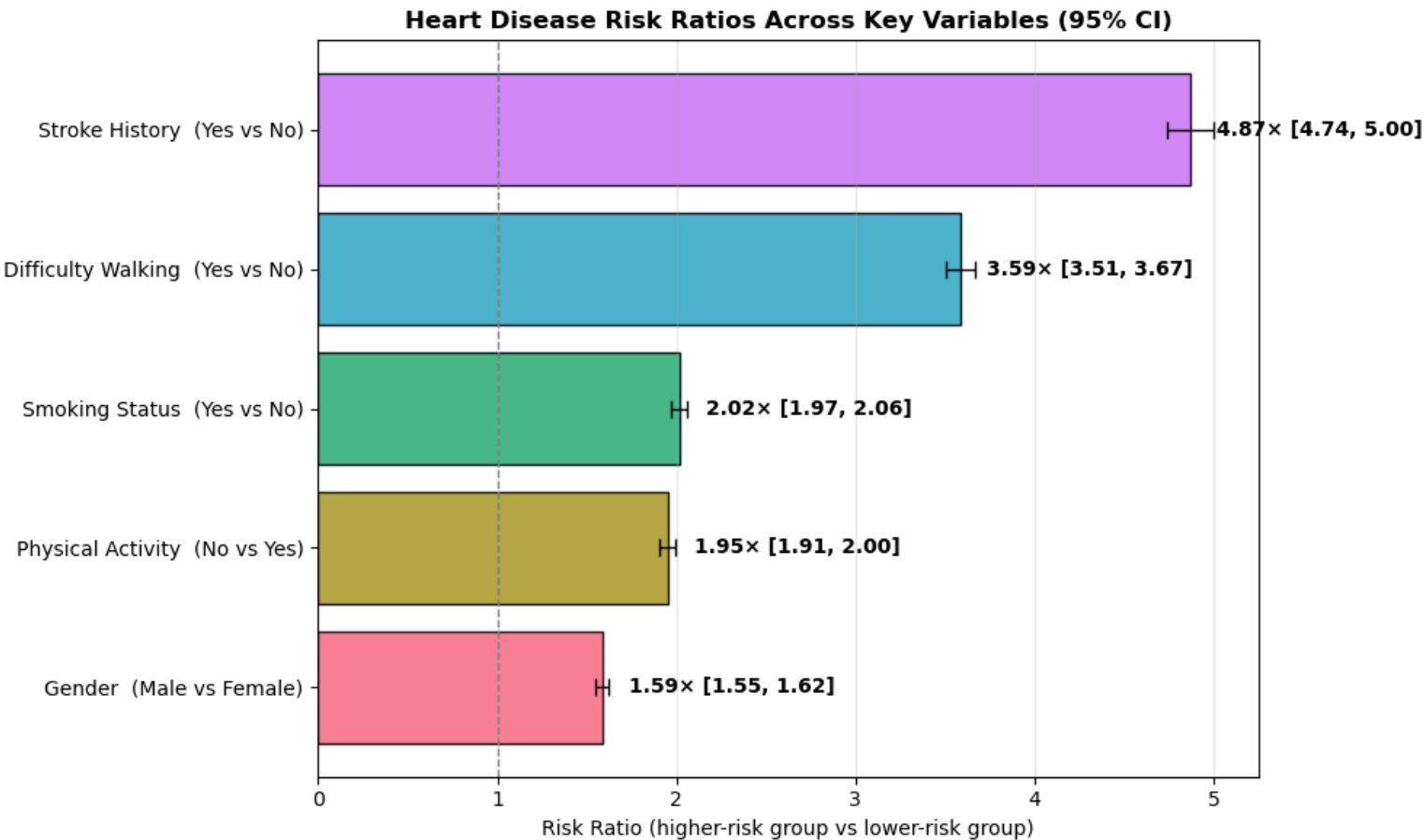
Heart disease rate: 22.6% vs 6.3%
3.6× higher risk



Smoking

Heart disease rate: 12.2% vs 6.0%
2.0× higher risk

📊 Risk Ratio Comparison



Analysis of numeric variables shows clear relationships with heart disease risk, with age and BMI exhibiting the strongest associations.



Age

Consistent increase in heart disease rates with advancing age, from 0.6% in younger adults to over 20% in older adults.



BMI

Obese category shows consistently highest heart disease rates (10.4%), followed by overweight (8.7%), normal weight (6.5%), and underweight (7.8%).



Physical Health

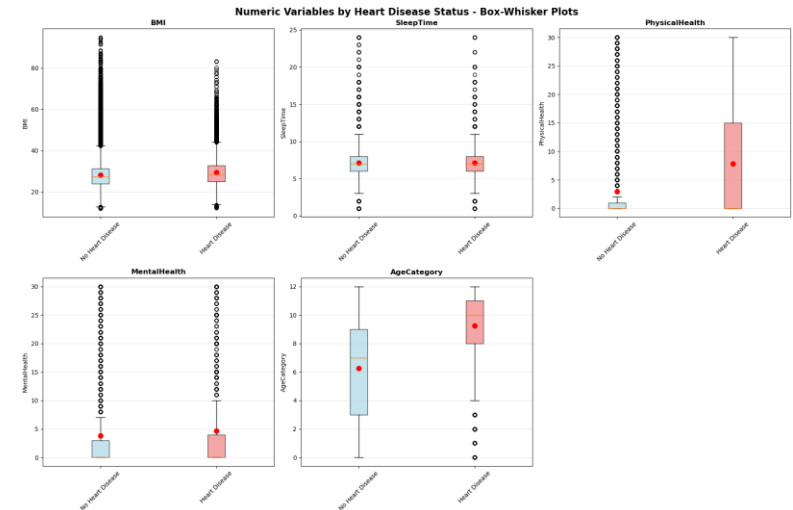
Poorer physical health scores correlate with higher heart disease rates, with a clear distinction between categories.



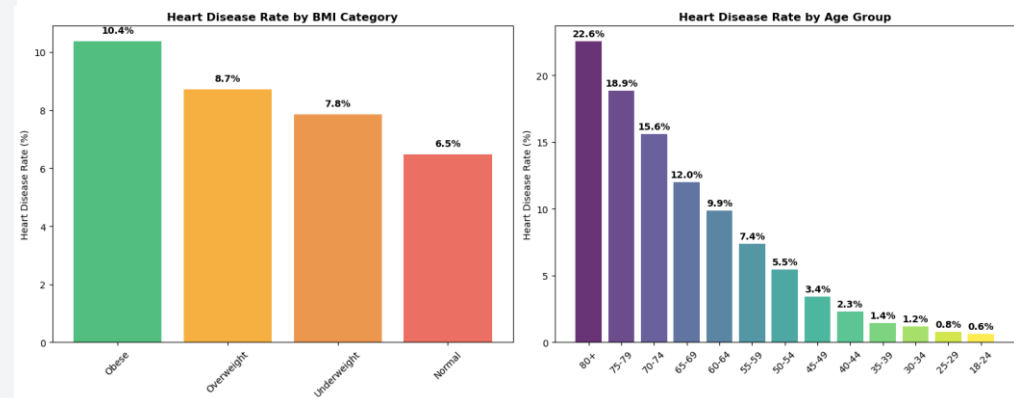
Mental Health

Some association with heart disease risk, though less pronounced than physical health factors.

Distribution of Health Metrics



Heart Disease Rates by Category



Key Insight: Age and BMI show the most consistent association with heart disease risk across the population.

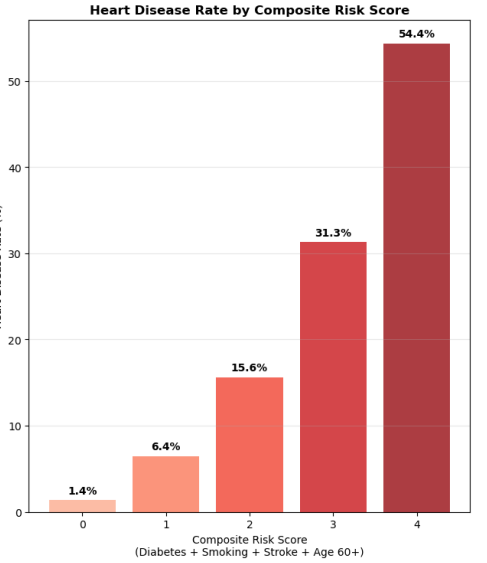
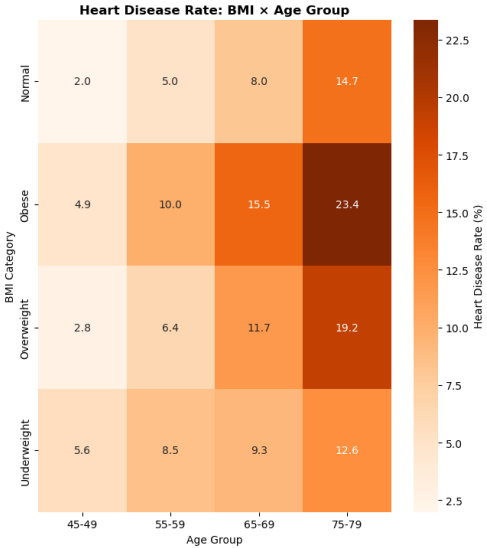
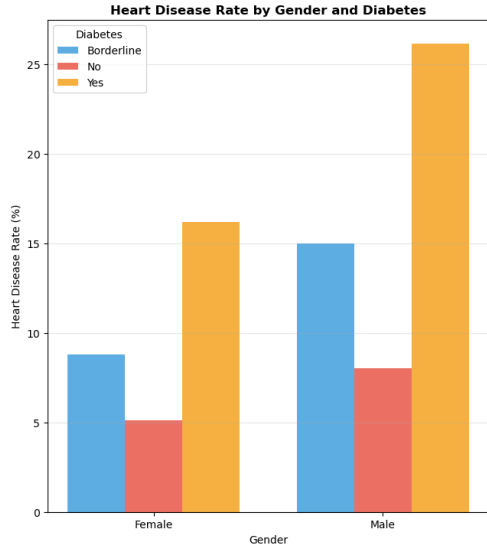
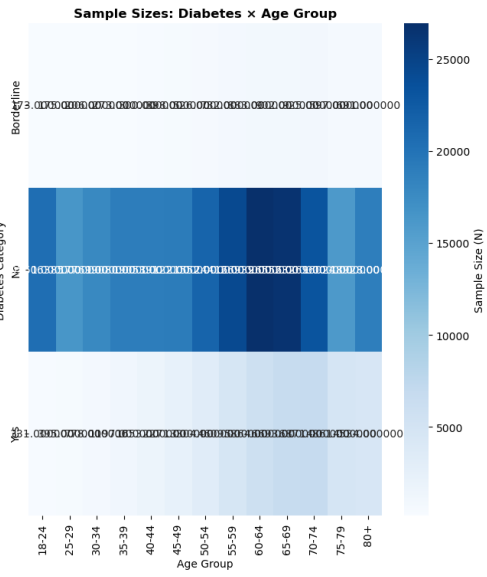
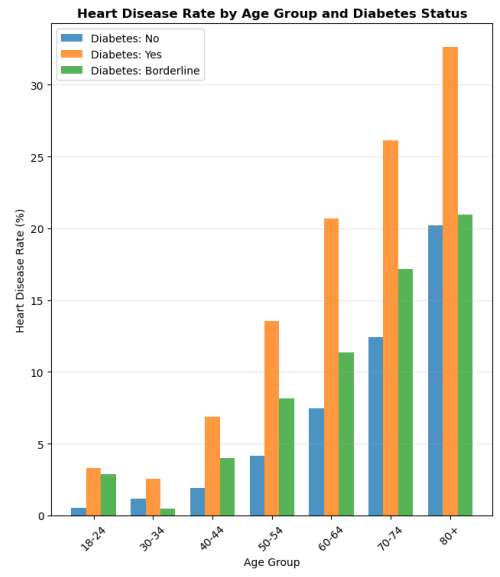
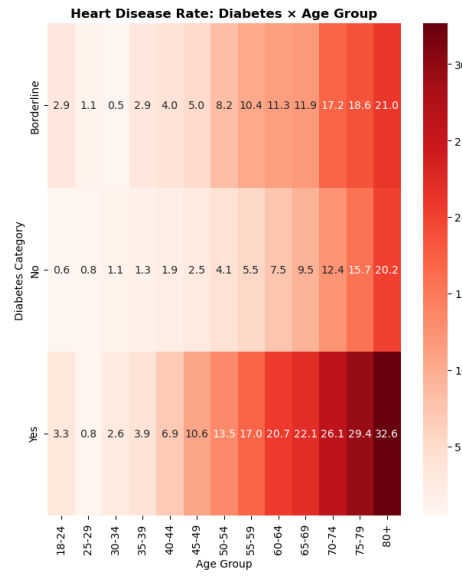
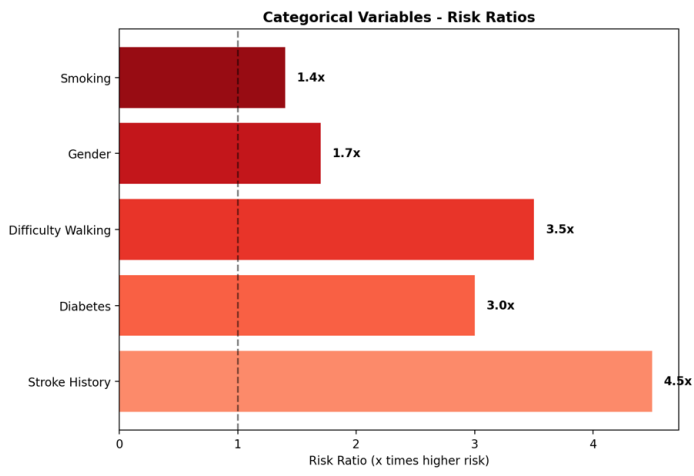
Analysis reveals that combinations of risk factors significantly amplify heart disease risk beyond individual factors.

Key Insights

- ✓ Individuals with 4+ risk factors have over 10x higher risk compared to those with no risk factors
- ✓ The combination of diabetes and smoking shows a compounding effect on risk
- ✓ Age 60+ combined with other factors creates a particularly high-risk profile

Composite Risk Score

A simple risk score was created by summing binary risk factors (Diabetes, Smoking, Stroke, Age 60+). The heart disease rate increased exponentially with this score, from ~3% with 0 risk factors to over 30% with 4 or more factors.



Model Development Methodology



Train-Test Split

80/20 split with stratification to maintain class proportions



Multicollinearity Assessment

Variance Inflation Factor (VIF) analysis identified SleepTime (VIF=14.87) as problematic



Threshold Optimization

Used Youden's Index to balance sensitivity (80.25%) and specificity (70.61%)



Model Evaluation

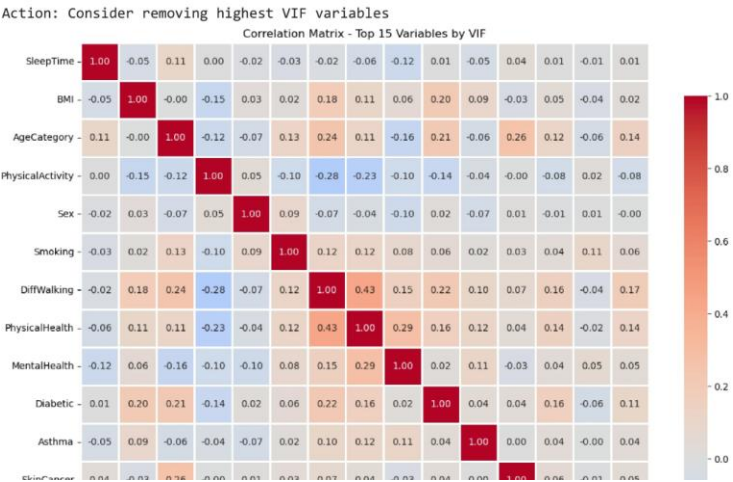
Comprehensive assessment using AUC (0.8291 training, 0.8262 test)



Key Insight

Random Forest emerged as the best-performing model with an AUC of **0.8269** on test data

Multicollinearity Assessment



Correlation matrix showing variables with high VIF values

Model Development Flow



Data Preparation



Feature Selection



Model Training



Deployment



Challenge Addressed

Class imbalance (8.56% heart disease cases) and multicollinearity

⚙️ Logistic Regression Methodology

- 80/20 train-test split (stratified)
- Removed multicollinearity (VIF=14.87 for SleepTime)
- 14 statistically significant predictors ($p < 0.001$)

⚖️ Threshold Optimization

Default threshold (0.5):

Sensitivity
9.12%

Specificity
99.20%

Optimal threshold (0.0782):

Sensitivity
80.25%

Specificity
70.61%

The initial high accuracy is misleading, driven by correctly classifying the majority "No Disease" class. The sensitivity of 9.12% is unacceptable for a medical screening tool.

Optimal Threshold Selection

Using Youden's Index (maximizing Sensitivity + Specificity), the optimal threshold was found to be **0.0782**.

📊 Model Performance

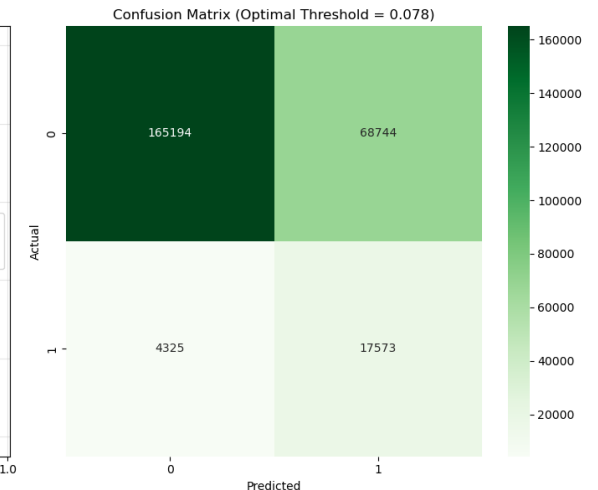
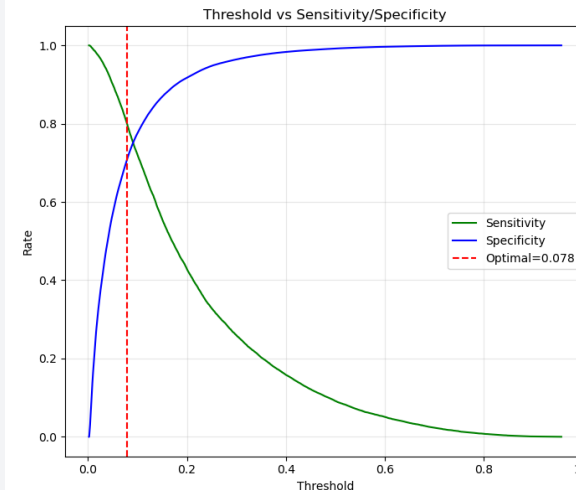
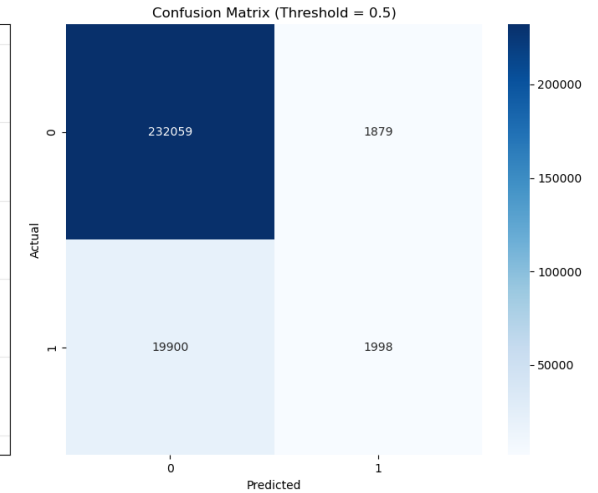
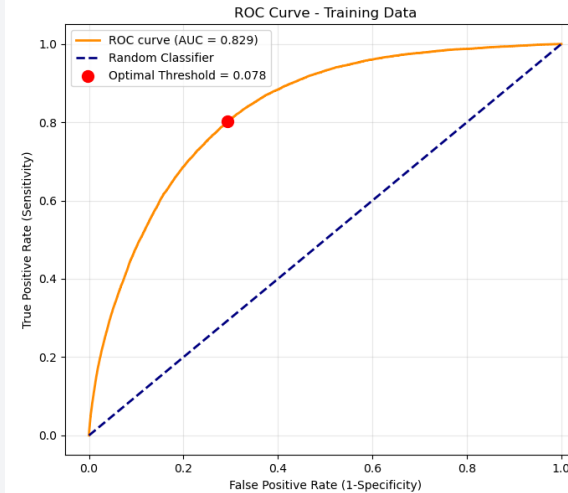
🎯 Training AUC
0.8291

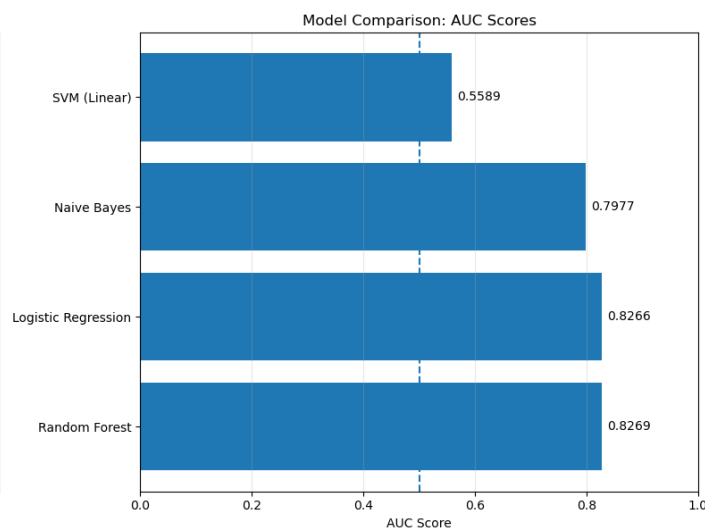
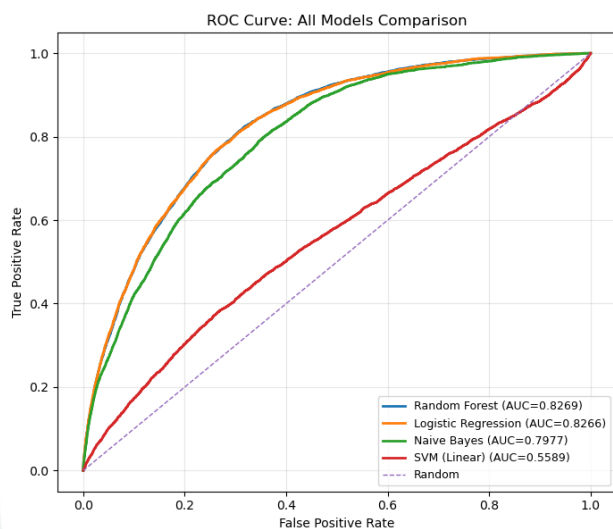
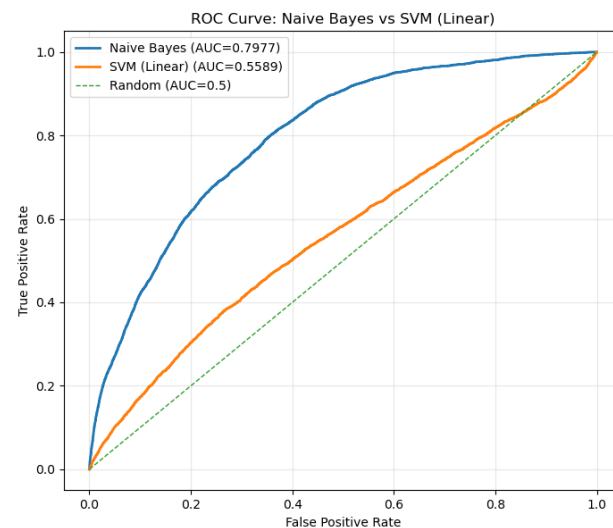
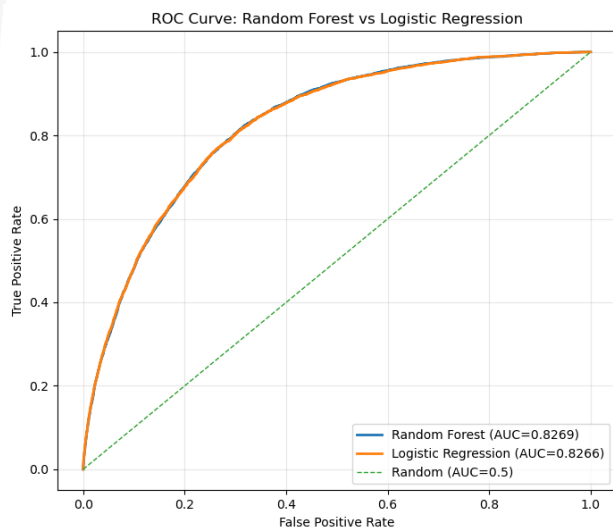
🎯 Test AUC
0.8262

✅ Accuracy
71.44%

↔ Overfitting
0.0324

📈 ROC Curve Analysis





Model Performance Comparison



Random Forest

AUC: 0.8269

Best overall performance with excellent balance across metrics.



Logistic Regression

AUC: 0.8266

Strong baseline model with interpretable results.



Naïve Bayes

AUC: 0.7977

Slightly lower performance but still effective.



SVM (Linear)

AUC: 0.5589

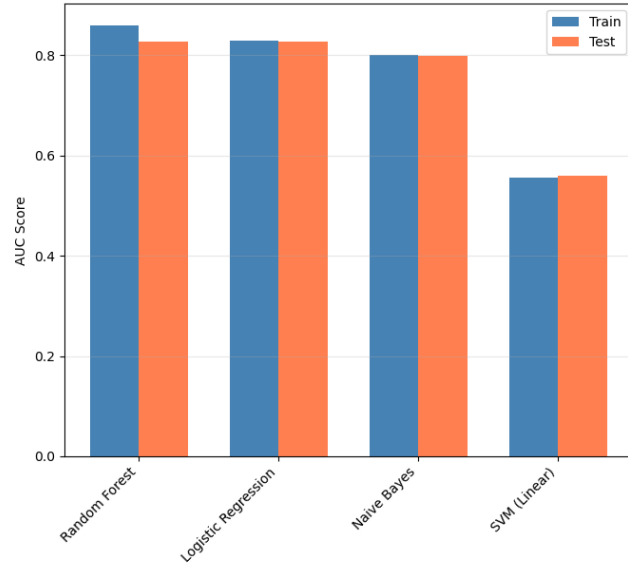
Competitive but slower training time.

ROC Curve Insights

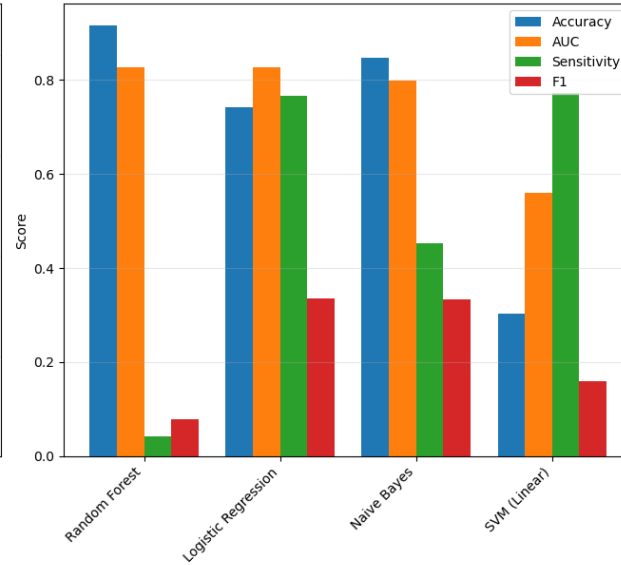
ROC curves illustrate the diagnostic ability of a binary classifier system as the discrimination threshold is varied. The area under the ROC curve (AUC) provides a single measure of overall classifier performance.

Model Performance Comparison

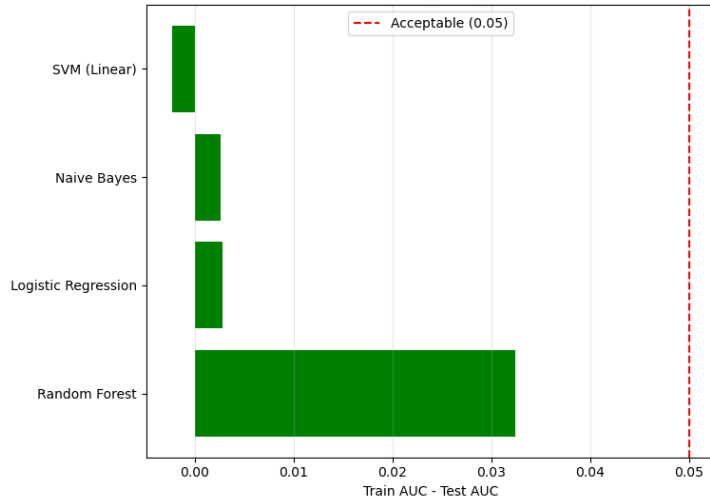
AUC Comparison: Train vs Test



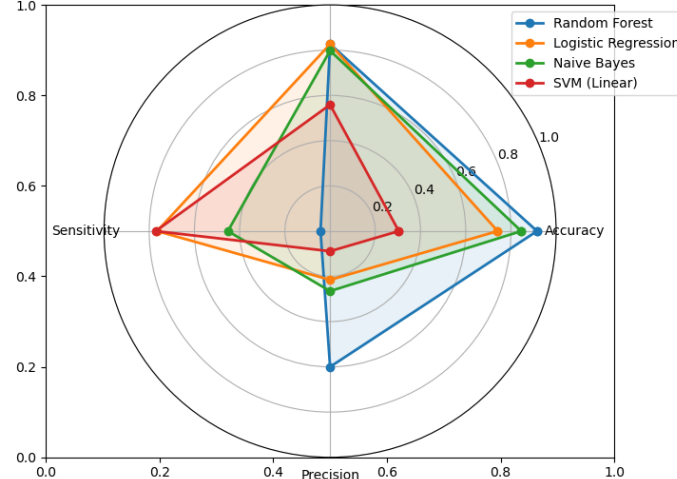
Test Metrics Comparison



Overfitting Analysis



Model Performance Radar



Best Model: Random Forest

Random Forest

BEST MODEL

Test AUC
0.8269

Overfitting
0.0324

Accuracy
0.9156

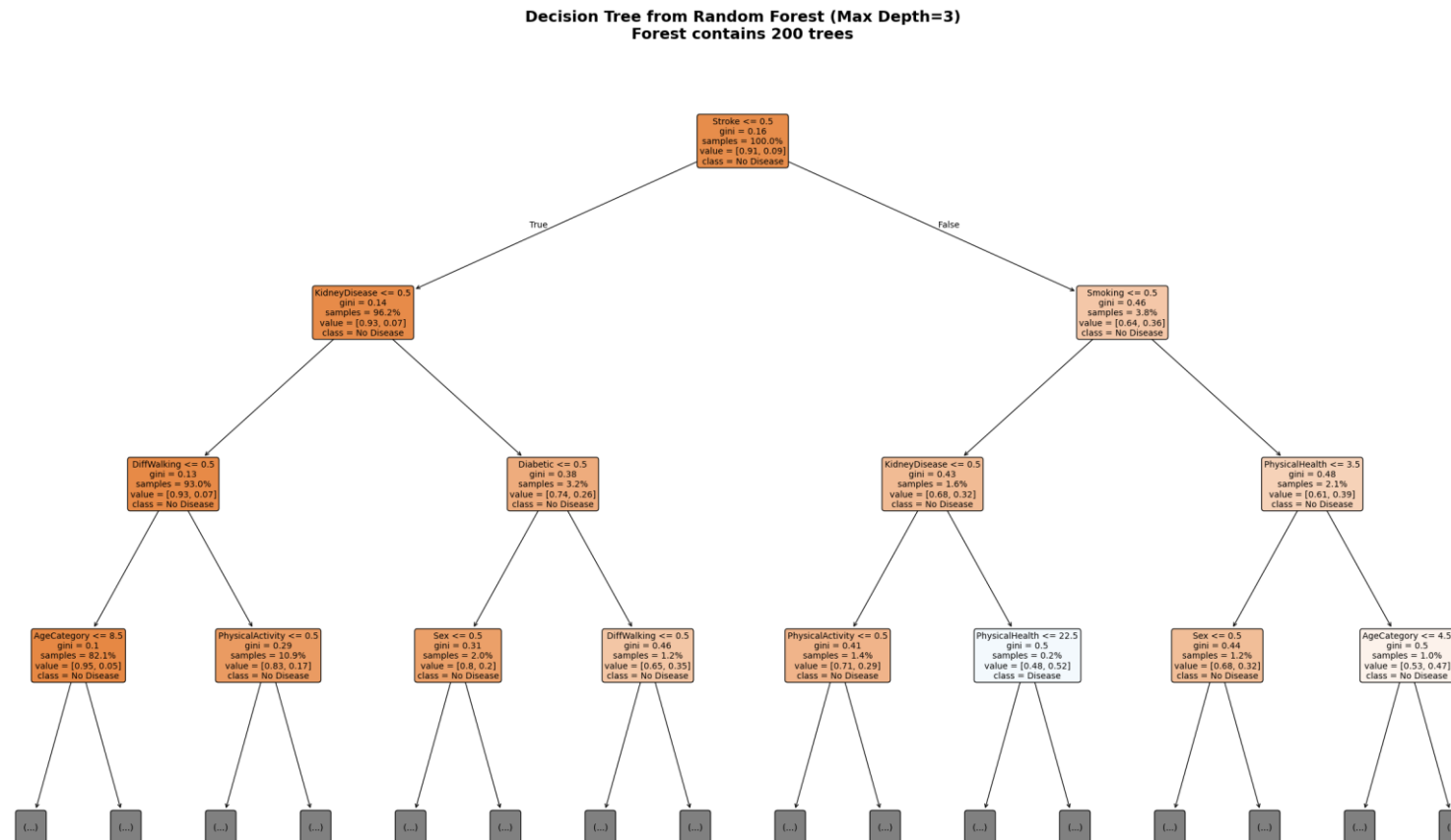
Sensitivity
0.0422

Selection Criteria

- Highest Test AUC: **Random Forest (0.8269)**
- Lowest Overfitting: **Random Forest (0.0324)**
- Balance of Accuracy, Sensitivity, and Precision

Reasons for Selection

Trade-off: The primary weakness of the Random Forest model is its very low sensitivity (4.22%) when using default settings. However, for a deployment scenario, the **probability scores** it generates can be used with the **optimized threshold (0.0782)** discovered during the Logistic Regression phase, which would restore high sensitivity. The Random Forest's strength lies in its superior ranking of instances (high AUC).



Key Findings

 **Stroke History:** 4.9x higher risk

 **Diabetes:** 3.2x increased risk

 **Gender:** Males have 1.6x higher risk

 **Physical Activity:** Protective factor

Best Model Performance: Random Forest

Test AUC
0.8269

Accuracy
0.9156

Precision
0.6000

Recommendations

Deployment Plan

- ✓ Deploy Random Forest model into a real-time prediction dashboard
- ✓ Used Streamlit for the dashboard implementation click [HERE](#)
- ✓ Target healthcare professionals for clinical decision support

Expected Benefits

-  Early intervention
-  Patient care
-  Improved outcomes
-  Resource optimization

🎯 Summary and Recommendations

This project successfully identified the key factors associated with heart disease and developed a high-performing predictive model.

- **Top Risk Factors:** Stroke History, Diabetes, Kidney Disease, Difficulty Walking, and Smoking.
- **Protective Factors:** Physical Activity and moderate Alcohol Consumption.
- **Demographics:** Risk increases significantly with age and is higher in males.
- **Compounding Risk:** Combinations of factors (e.g., Diabetes and Old Age) lead to exponentially higher risk.
- **Best Model:** The **Random Forest** algorithm provided the most powerful predictions (AUC: 0.8269).

STRATEGIC RECOMMENDATIONS

1. **Clinical Deployment:** Created and deployed the finalized Random Forest model into a user-friendly [Streamlit dashboard](#). This will allow healthcare professionals to input patient data and receive an immediate, evidence-based heart disease risk score.
2. **Preventive Health Focus:** Public health initiatives should prioritize managing conditions like diabetes and hypertension, and promoting physical activity, as these are major, modifiable risk factors.
3. **Screening Programs:** The model can be integrated into routine health screenings to flag high-risk patients for further diagnostic testing, enabling earlier intervention.

